

MOHAMMAD FARHAN

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SUMMARY

Computer Science (Computer Network) graduate from Universiti Putra Malaysia seeking a software engineering role. Experienced in building and deploying full-stack applications on cloud platforms, including AI-powered chatbot integrations. Skilled in API design, containerization (Docker), and system integration, with a strong foundation in networking (TCP/IP, DNS). Recognized for outstanding performance in regional technical competitions and real-world projects.

EDUCATIONAL BACKGROUND

Bachelor of Computer Science (Computer Network), Universiti Putra Malaysia	2021 - 2025
- CGPA: 3.96 - Vice Chancellor Awards, Dean List Awards	
Matriculation Programme, Kolej Matrikulasi Labuan	2020 - 2021
CGPA: 3.75	

WORK EXPERIENCE

Software Engineer Intern — Digital Heritage Sdn. Bhd.	Mar 2025 - Aug 2025
- Developed Python-based applications and conducting system testing to ensure functionality, reliability, and seamless integration. - Designed and deployed a microservice on DigitalOcean, integrating it with a Rocket.Chat chatbot via a custom plugin to automate response handling.	
Freelance Web Developer — Soloreen Ventures	Sept 2024 - Feb 2025
- Developed custom WordPress websites, focusing on performance, responsiveness, and user experience to improve usability and site reliability. - Collaborated with clients to gather requirements and translate business needs into scalable web solutions aligned with best practices.	

ACHIEVEMENTS

Huawei Developer Competition 2023 — Asia Pacific (Innovation Award)
- Developed a prototype of CommunityCARE, a complaint management system that enhance public engagement and transparency in local governance. - Integrated Huawei Cloud services, including Object Storage Service (OBS) for data storage and ModelArts for image recognition to support automated issue reporting.
APAC HPC-AI Competition 2023 (Merit Award)
- Optimized the Model for Prediction Across Scales (MPAS) by tuning processing elements and threading to improve simulation speed and resource utilization. - Developed PBS batch scripts to automate job scheduling and resource allocation, increasing throughput in distributed computing environments.

PROJECTS

PayTrack — Invoicing Platform

- Built a multi-tenant invoicing system with Java Spring Boot and Vue.js, featuring an AI-powered assistant integrated with Google Gemini.
- Deployed a serverless pipeline on AWS using Lambda function and EventBridge to pre-compute dashboard KPIs into S3 snapshots, reducing database load.

QRBridge — QR Code Generation Microservice

- Built a serverless Python microservice on DigitalOcean Functions exposing a REST API for QR code generation with configurable payloads and input validation.
- Integrated Rocket.Chat and the Telegram Bot API to support real-time QR code generation and delivery from chat commands.

CloudPulse — Uptime & Incident Monitoring Platform

- Built a service uptime monitoring system with scheduled health checks tracking status, latency, and response codes, with a real-time dashboard.
- Implemented a state-machine alert engine using FastAPI and APScheduler for consecutive failure detection, recovery tracking, and multi-channel notifications

DomainXray — DNS & WHOIS Analysis Tool

- Built a domain intelligence tool with a custom DNS resolver supporting 8 record types and a TCP WHOIS client with automatic referral-chain following across 80+ TLDs.
- Developed a FastAPI backend and Vue 3 + Tailwind frontend to consolidate DNS lookups, WHOIS queries, and registration data in a unified web interface.

AquaVision — Home-Based Aquaculture Monitoring System

- Built a React Native mobile app for real-time aquaculture monitoring, collecting environmental data via ESP32 microcontroller and sensors.
- Integrated Firebase Realtime Database and Cloud Functions to process sensor readings and trigger automated alert notifications.

SKILLS

- **Programming Languages:** Java, Python, JavaScript, Bash (Shell Scripting)
- **Backend:** Spring Boot, FastAPI, Node.js, RESTful APIs, Serverless Functions, LLM Integration
- **Frontend:** Vue.js, React Native, Tailwind CSS
- **Databases:** PostgreSQL, Firebase Realtime Database
- **Cloud & Infrastructure:** Docker, AWS, DigitalOcean, Huawei Cloud, Firebase
- **Networking:** TCP/IP, DNS, HTTP/HTTPS, TLS/SSL, Routing Protocols
- **Tools:** Git, GitHub, Linux (WSL)
- **Languages:** Malay (Native), English (Advanced)